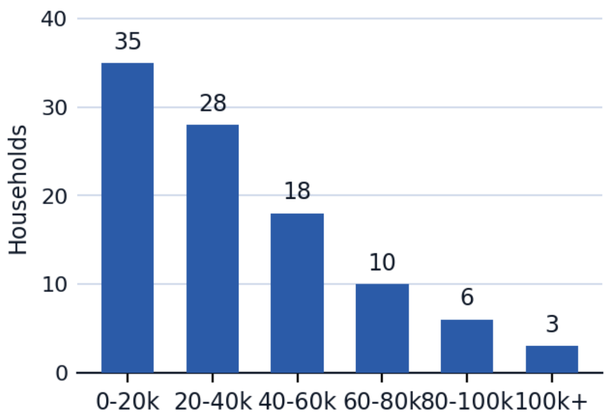
Describing Distributions & Linear Regression

Describing shape, center, and spread

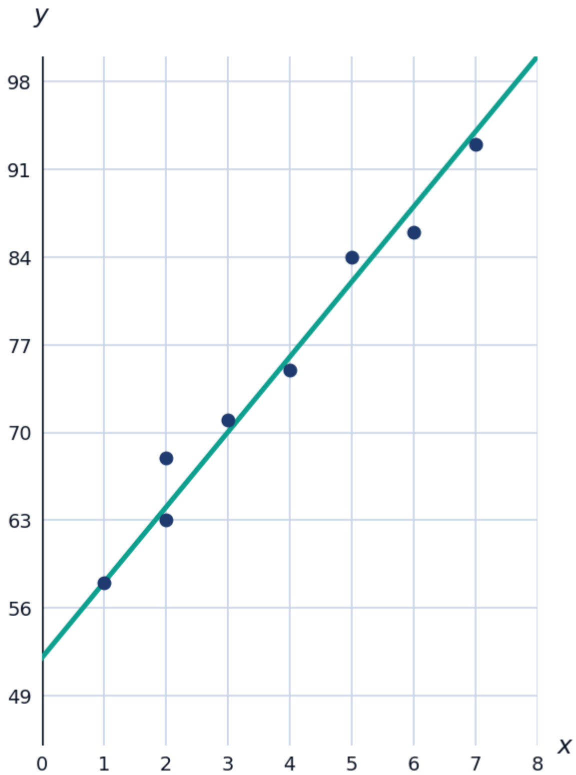
1 A histogram of household incomes has a long tail stretching toward high values, with most households clustered at lower incomes. Describe the shape of this distribution and state which measure of center (mean or median) is more appropriate to report, with justification.



2 Describe what it means for a distribution to be symmetric, and state the relationship between the mean and median in that case.

Scatterplots and correlation

3 The scatterplot shows hours studied (x) versus test score (y) for 8 students, along with the least-squares regression line $\hat{y} = 52 + 6x$.



a Based on the scatterplot, describe the direction and strength of the association between hours studied and test score.

b Use the regression equation to predict the test score for a student who studies 4.5 hours.

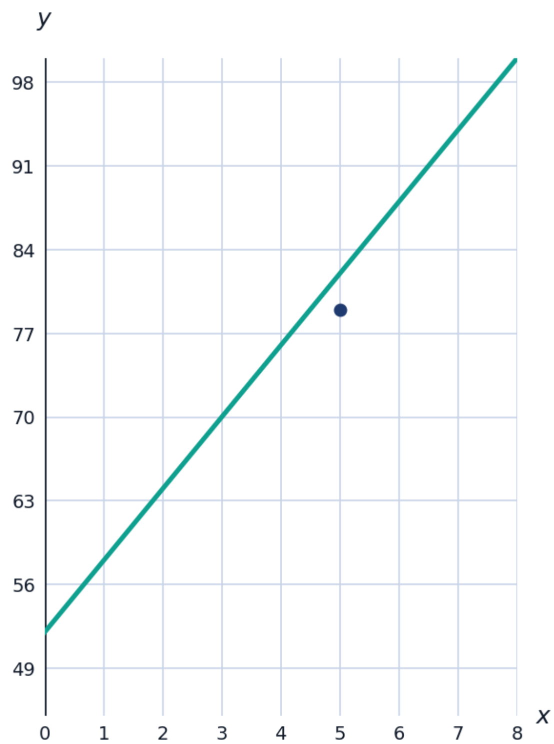
Interpreting the correlation coefficient

4 Explain what a correlation coefficient of $r = 0.85$ indicates about a linear relationship, and explain why r alone cannot confirm that x causes y.

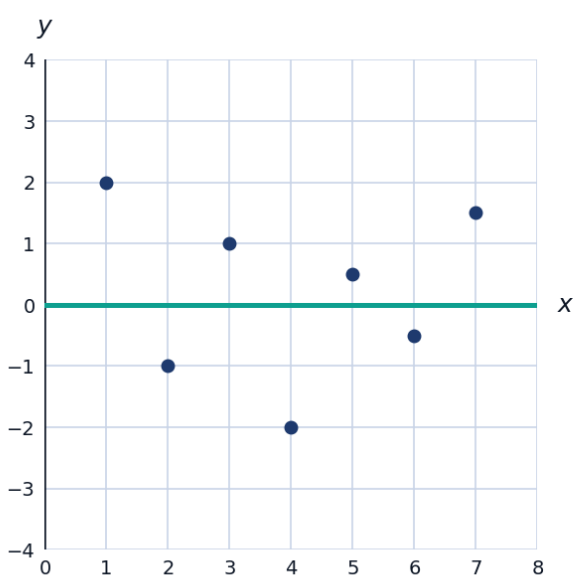
5 A data set has correlation coefficient $r = -0.4$. Describe the direction and strength of the linear relationship.

Least-squares regression and residuals

6 Using the regression equation $\hat{y} = 52 + 6x$ from earlier, find the residual for a student who studied 5 hours and actually scored 79.



7 Explain what it means for a residual plot to show "no clear pattern," and why this supports using a linear model for the data.



Coefficient of determination

8 For the study-hours regression, $r = 0.85$. Find r^2 and interpret its meaning in context.

Influential points and outliers in regression

9 A data point has an unusually large x-value far from the rest of the data, and removing it noticeably changes the slope of the regression line. Explain why this point is called influential, and contrast it with a point that has a large residual but does not change the line much when removed.

Extrapolation

10 Using the regression $\hat{y} = 52 + 6x$ (fit using data from students who studied between 1 and 7 hours), explain why predicting a test score for a student who studied 20 hours would be unreliable.